4.4 *Data Preparation*

The hierarchical CPI data is provided as monthly index values. We transformed these CPI values into monthly logarithmic change rates. Let x_t represent the CPI value (of any node) at month t. The logarithmic change rate at month t, denoted as $rate(t)$, is calculated as follows:

$$rate(t) = 100 \times \log\left(\frac{x_t}{x_{t-1}}\right). \tag{4.1}$$

Unless specified otherwise, the remainder of this paper focuses on monthly logarithmic change rates as defined in Equation (4.1).

We divided the data into a *training* set and a *test* set as follows: for each time series, the first 75% of the measurements (earliest in time) were assigned to the *training* set, while the remaining 25% were set aside for the *test* set. The *training* set was used to train the BiHRNN model and other baseline models. The *test* set was used for evaluation. The results presented in Section 5.4 are based on this data split.

4.4.1 *General Data Statistics*

Table 4.3 summarizes the number of data points and general statistics of the CPI time series after applying Equation (4.1). A comparison between the headline CPI and the full hierarchy reveals that lower levels exhibit significantly higher standard deviations (STD) and wider ranges, indicating greater volatility.

Table 4.3: Descriptive Statistics

Data Set	# Monthly Measurements	Mean	STD	Min	Max	# of Indexes	Avg. Measurements per Index
US - Headline Only	476	0.23	0.32	-1.93	1.31	1	476
US - Level 1	2064	0.19	0.85	-18.61	11.32	26	79.38
US - Level 2	7128	0.19	1.64	-32.92	16.67	24	297
US - Level 3	6476	0.17	1.75	-34.24	24.81	25	259.04
US - Level 4	9767	0.13	1.67	-35.00	28.17	48	203.48
US - Level 5	17656	0.11	2.39	-23.89	242.50	110	160.51
US - Level 6	10604	0.16	1.54	-16.49	17.06	73	145.26
US - Level 7	4968	0.21	1.64	-11.89	17.84	36	138
US - Level 8	980	0.21	1.70	-8.65	7.80	7	140

Data Set	# Monthly Measurements	Mean	STD	Min	Max	# of Indexes	Avg. Measurements per Index
Canada - Headline Only	121	0.20	0.40	-0.72	1.42	1	121
Canada - Level 1	2057	0.19	1.11	-10.78	8.17	17	121
Canada - Level 2	2541	0.18	1.56	-14.36	16.20	21	121
Canada - Level 3	7381	0.20	1.92	-20.69	27.92	61	121
Canada - Level 4	10648	0.17	2.29	-27.09	31.86	88	121
Canada - Level 5	9680	0.19	2.93	-34.03	35.43	80	121
Canada - Level 6	3025	0.20	2.48	-19.12	20.03	25	121

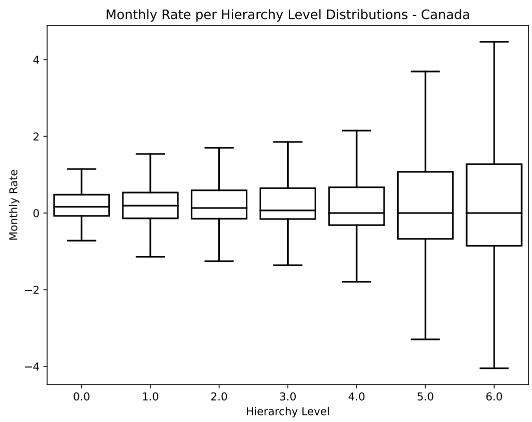
Data Set	# Monthly Measurements	Mean	STD	Min	Max	# of Indexes	Avg. Measurements per Index
Norway - Headline Only	172	0.22	0.45	-0.93	1.36	1	172
Norway - Level 1	2064	0.21	1.49	-12.82	10.88	12	172
Norway - Level 2	6708	0.21	2.15	-20.79	27.31	39	172

Notes: General statistics of the headline CPI and CPI per each level in the hierarchy across Canada, Norway, and the US.

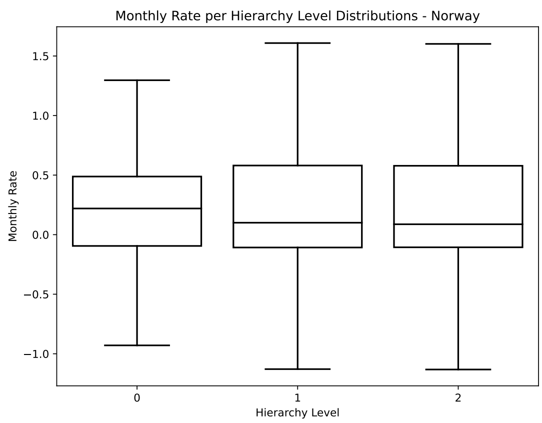
Figure 4.1 depicts box plots of the CPI change rate distributions at different levels. The boxes depict the median value and the upper 75'th and lower 25'th percentiles. The figure further emphasize that the change rates are more volatile as we go down the CPI hierarchy. A high dynamic range, high standard deviation, and limited training data all signal increased difficulty in making predictions within the hierarchy. Given this, we can anticipate that predictions for disaggregated components within the hierarchy will be more challenging than those for the headline.

Figure 4.1. ***CPI Distributions by Hierarchy Level for Canada, Norway, and the US***

(a) *Canada CPI by Hierarchy Level*



(b) *Norway CPI by Hierarchy Level*



(c) *US CPI by Hierarchy Level*